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INFORMATION PROCESSING DEVICE, IMAGE FORMING APPARATUS, CONTROL METHOD THEREOF, AND STORAGE MEDIUM

Abstract

In a case where a user instructs an image forming apparatus to perform processing of sending scanned image data to the user from a mobile terminal, the user needs to input a user's e-mail address on the mobile terminal, which includes an inconvenient operation. From the mobile terminal, the user instructs the image forming apparatus to send the data to the user's e-mail address which is preliminarily registered in the image forming apparatus.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 18/054,078, filed Nov. 9, 2022, which is a Continuation of U.S. patent application Ser. No. 17/148,389, filed Jan. 13, 2021, now U.S. Pat. No. 11,516,368, which claims the benefit of Japanese Patent Application No. 2020-014214, filed Jan. 30, 2020, all of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

Field of the Disclosure

[0002] The present disclosure relates to a data processing device including a sending unit, an information processing method of the data processing device, and a storage medium.

Description of the Related Art

[0003] In recent years, a high-performance mobile terminal called a smartphone has become widespread. The mobile terminal can send instructions for scanning, printing, and the like to an image forming apparatus, and operate the image forming apparatus. For example, an operation to send image data scanned by the image forming apparatus to a set destination can be instructed from the mobile terminal. The mobile terminal configures a job by adding an e-mail address as a sending destination of image data set on the mobile terminal by a user and various kinds of scan setting information to the job. The mobile terminal then sends the job to the image forming apparatus. The image forming apparatus, which has received the job, sends image data scanned in accordance with a scan setting included in the job to the e-mail address as the sending destination included in the job. In this manner, a series of setting operations necessary for sending the scanned image data can be completed using the mobile terminal. In this operation, communication between the mobile terminal and the image forming apparatus is established in accordance with a communication protocol, such as Transmission Control Protocol/Internet Protocol (TCP/IP), Near Field Communication (NFC), and Bluetooth®. In recent years, after the user performs settings on the mobile terminal, the user can also operate the image forming apparatus in accordance with the settings via the NFC or the like by simply bringing the mobile terminal close to (in touch with) the image forming apparatus.

[0004] Meanwhile, after a scan setting and a sending destination setting are performed on an operation panel of the image forming apparatus, the image forming apparatus can also send scanned image data. As a method of easily selecting an e-mail address of a user who is performing an operation at the time of setting a sending destination on the operation panel of the image forming apparatus, Japanese Patent Application Laid-Open No. 2019-68336 proposes the following method. That is, a user who can log in to the image forming apparatus and the user's e-mail address are preliminarily registered in association with each other in the image forming apparatus. When the user logs in to the image forming apparatus, the image forming apparatus displays a send-to-myself button to set the log-in user's e-mail address as a destination to which scanned image data or the like is sent. Japanese Patent Application Laid-Open No. 2019-68336 also discusses a method in which the image forming apparatus calls the user's e-mail address registered in association with the user when the user selects the send-to-myself button and sets the e-mail address as the destination.

[0005] In a case where the user instructs the image forming apparatus to perform processing of

sending scanned image to the user from the mobile terminal, the user needs to input the user's e-mail address using the mobile terminal, which requires the user to perform an inconvenient operation.

SUMMARY

[0006] Embodiments of the present disclosure are directed to increasing user operability in a case where the user instructs the image forming apparatus to send data to the user from the mobile terminal. According to embodiments of the present disclosure, a method for an information processing device capable of communicating with an image forming apparatus, the method includes sending first information including user information of a user operating the information processing device and information to instruct the image forming apparatus to send image data to a destination of the user which is preliminarily registered in the image forming apparatus in correspondence with the user information, to the image forming apparatus.

[0007] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram illustrating a system configuration according to an exemplary embodiment.

[0009] FIG. 2 is a diagram illustrating a hardware configuration of a data processing device according to the present exemplary embodiment.

[0010] FIG. 3 is a diagram illustrating a hardware configuration of an image forming apparatus according to the present exemplary embodiment.

[0011] FIG. 4 is a block diagram illustrating a software configuration of the data processing device according to the present exemplary embodiment.

[0012] FIG. 5 is a block diagram illustrating a software configuration of the image forming apparatus according to the present exemplary embodiment.

[0013] FIG. 6 is a diagram illustrating an example of a sending setting screen of the image forming apparatus according to the present exemplary embodiment.

[0014] FIG. 7 is a diagram illustrating an example of a menu screen of the data processing device according to the present exemplary embodiment.

[0015] FIG. 8 is a diagram illustrating an example of a send-to-myself screen of the data processing device according to the present exemplary embodiment.

[0016] FIG. 9 is a diagram illustrating an example of a send-to-myself screen of the data processing device according to the present exemplary embodiment.

[0017] FIG. 10 is a diagram illustrating an example of a color setting screen of the data processing device according to the present exemplary embodiment.

[0018] FIG. 11 is a flowchart describing processing for executing a send-to-myself operation in the data processing device according to the present exemplary embodiment.

[0019] FIG. 12 is a flowchart describing processing for executing a send-to-myself operation in the image forming apparatus according to the present exemplary embodiment.

[0020] FIG. 13 is a flowchart describing processing for resetting an e-mail address in the image forming apparatus according to the present exemplary embodiment.

[0021] FIG. 14 is a flowchart describing processing for resetting an e-mail address in the data processing device according to the present exemplary embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0022] Exemplary embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings. Note that constituent elements described in the exemplary

embodiments are merely examples, and are not intended to limit the scope of the present disclosure.

[0023] FIG. 1 is a block diagram illustrating a configuration of a data processing system according to a first exemplary embodiment. A data processing system **100** includes a data processing device **101**, a wireless local area network (LAN) terminal **102**, an image forming apparatus **104**, and a cloud server **105**, which are connected with one another via a network **103**, such as a LAN.

[0024] The data processing device **101** is, for example, a mobile terminal, such as a smartphone, and may be an information processing device in which an operating system for a compact terminal and a program for controlling an audio phone call, positional detection, and data communication operate. Alternatively, the data processing device **101** may be a personal computer not having functions, such as audio phone call control and positional detection control. The data processing device **101** is connected with the network **103** by the wireless LAN terminal **102**.

[0025] The wireless LAN terminal **102** is a wireless LAN master unit having a typical network router function, and provides a wireless LAN in, for example, a home and an office.

[0026] The image forming apparatus **104** is a multi-function peripheral having various kinds of functions, such as a printer function, a copy function, a scan function, and a fax function. The image forming apparatus **104** may be configured to incorporate a wireless access point, and be capable of directly communicating with the data processing device **101**. From the data processing device **101** in communication with the image forming apparatus **104**, a user can instruct the image forming apparatus **104** to execute a job to cause the image forming apparatus **104** to execute a function of the image forming apparatus **104**.

[0027] The cloud server **105** exchanges data with the data processing device **101** or the image forming apparatus **104** via the network **103**. The cloud server **105**, for example, manages data that is used in the data processing device **101** and the image forming apparatus **104** and performs processing of, for example, extending various kinds of functions. While, in the present exemplary embodiment, the image forming apparatus **104** and the cloud server **105** have wired connection with the network **103**, a wireless connection may be used using the wireless LAN terminal **102** similarly to the data processing device **101**.

[0028] The data processing device **101** and the image forming apparatus **104** are also capable of performing short-range wireless communication using Near Field Communication (NFC), Bluetooth® Low Energy (Bluetooth® LE), or the like. The image forming apparatus **104** sends connection information (such as an Internet Protocol (IP) address, a media access control (MAC) address, and subsystem identification (SSID)) to allow the data processing device **101** to have wireless LAN connection with the image forming apparatus **104** via an NFC communication unit or a Bluetooth® LE communication unit, which will be described below. The data processing device **101** then starts wireless LAN communication with the image forming apparatus **104** using the acquired connection information.

[0029] FIG. 2 is a diagram illustrating a hardware configuration of the data processing device **101** according to the present exemplary embodiment. The data processing device **101** includes a controller unit **201**. The controller unit **201** controls various kinds of communication units, i.e., an NFC communication unit **210**, a Bluetooth® LE communication unit **211**, and a wireless LAN communication unit **212**, and various kinds of user interface (UI) units, i.e., a microphone/speaker **213**, a display **214**, and a touch panel **215**.

[0030] The controller unit **201** includes a central processing unit (CPU) **202**, a read-only memory (ROM) **203**, a random-access memory (RAM) **204**, a network interface (I/F) **205**, an audio control unit **206**, a display control unit **207**, an input control unit **208**, and a storage device **209**, which are connected with one another by a system bus **216**.

[0031] The CPU **202** controls the whole system of the data processing device **101**. An operating system of the data processing device **101** and a program for controlling a phone call, data communication, and the like are stored in the ROM **203**, and the CPU **202** executes various kinds

of programs. The RAM **204** is used as a main memory of the CPU **202** and a temporary storage area of a work area or the like.

[0032] The storage device **209** is a nonvolatile storage device, and records settings for various kinds of operation modes, operating logs, and the like that are held in the storage device **209** even after a restart of the data processing device **101**.

[0033] The network I/F **205** is connected with the NFC communication unit **210**, the Bluetooth® LE communication unit **211**, and the wireless LAN communication unit **212**, and exchanges data with the image forming apparatus **104** or the cloud server **105** via various kinds of wireless communication.

[0034] The audio control unit **206** performs input/output control of audio data via the microphone/speaker **213**. The display control unit **207** performs output control of a screen displayed on the display **214**. The input control unit **208** performs input control of information instructed by the user via a button or the touch panel **215**. Various kinds of applications executed in the data processing device **101** use the audio control unit **206**, the display control unit **207**, the input control unit **208**, and the like.

[0035] FIG. **3** is a diagram illustrating a hardware configuration of the image forming apparatus **104** according to the present exemplary embodiment. The image forming apparatus **104** includes a controller unit **301** that controls various kinds of communication units, i.e., an NFC communication unit **310**, a Bluetooth® LE communication unit **311**, and a wireless LAN communication unit **312**, an operation unit **307**, a scanner **313**, and a printer **314**. In a case where the user uses a copy function of the image forming apparatus **104**, the controller unit **301** controls the scanner **313** to acquire image data of an original document, and controls the printer **314** to output an image on paper. In a case where the user uses a scan and send function, the controller unit **301** controls the scanner **313** to acquire image data of an original document and converts the image data to coded data. The controller unit **301** then sends the coded data to the data processing device **101**, the cloud server **105**, or the like via a network I/F **308**. In a case where the user uses a print function, the controller unit **301** receives the image data (coded data) from the data processing device **101** or the cloud server **105** via the network I/F **308**. The controller unit **301** then sends the received image data to the printer **314**. The printer **314** prints an image on paper as an output on the basis of the received image data. The image forming apparatus **104** also has a fax receiving function to receive data from an integrated services digital network (ISDN) or the like and print the data, and a fax sending function to send scanned data to the ISDN or the like. Work to be processed according to the setting or the like set by the user to execute each function described above is referred to as a job, and the image forming apparatus **104** executes predetermined processing in accordance with the job.

[0036] The controller unit **301** includes a CPU **302**, a RAM **303**, a ROM **304**, a hard disk drive (HDD) **305**, an operation unit I/F **306**, the network I/F **308**, and a device I/F **309**, which are connected with one another by a system bus **315**.

[0037] The CPU **302** controls the whole system of the image forming apparatus **104**. The RAM **303** is a system work memory for the CPU **302** to operate, and image data or the like is temporarily stored in the RAM **303**. In addition, scanned image data read by the scanner **313**, print image data received from the data processing device **101** via the network **103**, or the like is stored in the RAM **303**. The system's boot program, an application, and the like are stored in the ROM **304**. An operating system, system software, application software, image data, setting data, and the like are stored in the HDD **305**.

[0038] The operation unit I/F **306** outputs information to be displayed on the operation unit **307** to the operation unit **307**. The operation unit I/F **306** also accepts information input by the user from the operation unit **307**. The network I/F **308** is connected with the NFC communication unit **310**, the Bluetooth® LE communication unit **311**, and the wireless LAN communication unit **312**, and exchanges data with the data processing device **101** and the cloud server **105** via various kinds of

wireless communication. The wireless LAN communication unit **312** performs wireless LAN communication with the data processing device **101** via the network **103**. The NFC communication unit **310** and the Bluetooth® LE communication unit **311** perform short-range wireless communication with the data processing device **101**. The image forming apparatus **104** receives job setting information, image data, a job execution command, and the like from the data processing device **101**, and executes a job. The device I/F **309** connects the scanner **313** and the printer **314** for reading image data and executing print with the controller unit **301**, and inputs/outputs the image data.

[0039] FIG. **4** is a diagram illustrating a software configuration of the data processing device **101** according to the present exemplary embodiment. FIG. **4** is a functional block diagram of software implemented by the CPU **202** reading a control program stored in the ROM **203** or the storage device **209**.

[0040] An application **401** is an application installed in the data processing device **101**. The user performs setting on the application **401** to create a job, and the job can be sent to the image forming apparatus **104**. While various applications other than the application **401** are installed in the data processing device **101**, a description thereof is omitted. A configuration of the application **401** will be described below.

[0041] A UI control unit **402** accepts job setting information or the like input to the touch panel **215** by the user, and sends an accepted content to a job setting management unit **406** via the input control unit **208**. The UI control unit **402** also receives a response from the job setting management unit **406** or the job control unit **407**, and outputs a screen in accordance with the received content on the display **214** via the display control unit **207**. The UI control unit **402** yet also accepts user information input to the touch panel **215** by the user and authentication information for a log-in to the image forming apparatus **104**, such as a password, and sends the accepted information to an authentication setting management unit **403** via the input control unit **208**.

[0042] The authentication setting management unit **403** registers therein the authentication information received from the UI control unit **402**. The authentication setting management unit **403** then performs processing for the log-in to the image forming apparatus **104** on the basis of the authentication information. Specifically, the authentication setting management unit **403** hands off the authentication information to a network control unit **405**. The authentication information is sent to the image forming apparatus **104** via the network **103**. The image forming apparatus **104** performs log-in processing using the received authentication information, and returns a result to the data processing device **101**.

[0043] A device connection unit **404** establishes short-range wireless communication using the NFC or the Bluetooth® LE or wireless LAN communication via the network **103** between the data processing device **101** and the image forming apparatus **104**. For example, in the case of the NFC, the short-range wireless communication is started by the user bringing the NFC communication unit **210** of the data processing device **101** close to the NFC communication unit **310** of the image forming apparatus **104**. In the case of the Bluetooth® LE, the Bluetooth® LE communication unit **311** of the image forming apparatus **104** sends a Bluetooth® LE beacon, and the Bluetooth® LE communication unit **211** of the data processing device **101** receives the sent beacon. In a case where the device connection unit **404** determines that a radio field intensity of the beacon received by the data processing device **101** reaches a predetermined intensity or higher by the user bringing the data processing device **101** close to the image forming apparatus **104**, the short-range wireless communication is started. In this manner, the device connection unit **404** performs the short-range wireless communication with the image forming apparatus **104** by the user bringing the NFC communication unit **210** of the data processing device **101** close to the NFC communication unit **310** of the image forming apparatus **104** or bringing the Bluetooth® LE communication unit **211** of the data processing device **101** close to the Bluetooth® LE communication unit **311** of the image forming apparatus **104**. The device connection unit **404** acquires device information including

connection information to be used for wireless LAN communication from the image forming apparatus **104** via the short-range wireless communication. The device connection unit **404** starts wireless LAN communication with the image forming apparatus **104** via the network **103** using the acquired connection information.

[0044] The network control unit **405** sends job setting information, a job execution command, image data, user authentication information, and the like to the image forming apparatus **104** via the network **103**. In a case of configuring a job to send scanned image data to a set destination, the job setting information is setting information regarding a destination to which data is sent, a scan setting, and the like. The network control unit **405** also receives the job setting information and the like stored in the image forming apparatus **104** via the network **103**.

[0045] The job setting management unit **406** stores job setting information input by the user and accepted by the UI control unit **402** and job setting information received by the network control unit **405** from the image forming apparatus **104** or the cloud server **105** in the RAM **204** or the storage device **209**. In this operation, the job setting management unit **406** may register these pieces of job setting information as “favorites” as settings frequently used by the user. In addition, the job setting management unit **406** may register predetermined setting information as “preset” at the time of installation of the application **401**.

[0046] The job control unit **407** generates a job from the job setting information included in the “favorites” or “preset” registered in the job setting management unit **406** or job setting information set by a user operation, and sends a job execution instruction to the image forming apparatus **104** via the network **103**. In addition, the job control unit **407** also acquires a job execution situation and a device operation status of the scanner **313** or printer **314** from the image forming apparatus **104** via the network **103**, and sends the job execution situation and the device operation status to the UI control unit **402**.

[0047] FIG. **5** is a diagram illustrating a software configuration of the image forming apparatus **104** according to the present exemplary embodiment. FIG. **5** is a functional block diagram of software implemented by the CPU **302** reading a control program stored in the ROM **304** or the HDD **305**.

[0048] The application management unit **501** manages an application that operates on the image forming apparatus **104**. A UI control unit **502** accepts job setting information and the like input by the user via the operation unit **307**, and hands off the job setting information to a job setting management unit **506**. In addition, the UI control unit **502** receives a response from the job setting management unit **506** or a job control unit **507**, and outputs a screen in accordance with the response to the operation unit **307**.

[0049] An authentication unit **503** performs log-in processing on the basis of authentication information received from the data processing device **101**, and returns a result. The image forming apparatus **104** holds various kinds of information such as an e-mail address corresponding to a log-in user.

[0050] A device connection unit **504** establishes wireless LAN communication via short-range wireless communication with the data processing device **101**. A network control unit **505** receives job setting information, a job execution instruction command, image data, user authentication information, and the like from the data processing device **101** via the network **103**. The network control unit **505** also sends job setting information stored in the image forming apparatus **104** to the data processing device **101** via the network **103**.

[0051] In the RAM **303** or the HDD **305**, the job setting management unit **506** stores job setting information input by the user and accepted by the UI control unit **502** and job information including job setting information received from the data processing device **101** or the cloud server **105** by the network control unit **505**. In this operation, the job setting management unit **506** may register these pieces of job setting information as “favorites” as settings frequently used by the user. In addition, the job setting management unit **506** may register predetermined setting information as “preset”. The job setting management unit **506** also manages jobs executed in the image forming

apparatus **104** as “history”.

[0052] The job control unit **507** controls units including a scan processing unit **511**, a copy processing unit **512**, a print processing unit **513**, and a fax processing unit **514** in accordance with the received job information and executes various kinds of jobs. In this operation, the job control unit **507** executes various kinds of jobs using job setting information set by the user on the operation unit **307** and job setting information registered as the “favorites” or the “preset” by the job setting management unit **506**. The job control unit **507** also receives a job execution request received from the data processing device **101** or the like via the network **103**, and executes various kinds of jobs in accordance with the received job setting information and job execution instruction. The job control unit **507** yet also sends a job execution situation and a device operation status of the scanner **313** or printer **314** to the data processing device **101** via the network **103**.

[0053] FIG. **6** illustrates an example of a setting screen to send scanned image data to a set destination. The setting screen is displayed on the image forming apparatus **104**. The setting screen is a screen that is displayed by the user calling an application to perform a scan and send operation on the image forming apparatus **104**.

[0054] A screen **601** is a screen to set a sending destination. On the screen **601**, the sending destination can be selected and a scan setting can be performed. A send-to-myself button **602** is a button to add an e-mail address corresponding to a log-in user as a sending destination by being pressed in a case where the e-mail address corresponding to the log-in user is preliminarily registered in the image forming apparatus **104**. While, in the present exemplary embodiment, an administrator of the image forming apparatus **104** can register a list that associates identification (ID) of each user who can log in to the image forming apparatus **104** with an e-mail address of each user, the configuration is not limited thereto. For example, each user may be able to preliminarily register and change his/her own e-mail address. A save-to-my-folder button **603** is a button to add, in a case where the log-in user registers a specific destination to save (my folder), my folder as a sending destination. In a case where my folder has yet to be registered, the destination for saving can be set and registered by pressing the save-to-my-folder button **603**. An e-mail address as a sending destination and the like are displayed in a destination list **604**.

[0055] A description will be given of selection of a function of the application **401** on the data processing device **101** with reference to FIG. **7**. FIG. **7** illustrates a menu screen of the application **401** that operates on the data processing device **101**. The application **401** can communicate with the image forming apparatus **104** and issue an instruction to execute processing based on contents set on the application **401** to the image forming apparatus **104**.

[0056] On a menu screen **703** of the application **401**, the UI control unit **402** displays a list of processing that uses functions of the image forming apparatus **104**. The displayed list includes jobs registered as, for example, the “favorites” and the “preset” described above. The list displays, for example, a send-to-myself button **701** to send scanned image to a user (who has logged in to the image forming apparatus **104**) by e-mail, and a save-to-my-folder button **702** to save the scanned image to a user's registered folder. In a case where the send-to-myself button **701** is pressed, the screen transits to a send-to-myself setting screen **710** to send an e-mail to the user. In a case where the save-to-my-folder button **702** is pressed, the screen transits to a setting screen (not illustrated) to save the scanned image to my folder. A plurality of buttons to execute jobs that use basic functions of the image forming apparatus **104**, such as a “copy” button and a “print” button is arranged in a basic job button area **704**. In a case where any of the buttons is pressed, the screen transits to a corresponding setting screen to execute a job.

[0057] Before referring to FIGS. **8** to **10**, an overview of the present exemplary embodiment will be described. In a case where the user presses the send-to-myself button **701**, performs a setting on the setting screen, and brings the data processing device **101** in touch with the image forming apparatus **104**, the user can send image data scanned by the image forming apparatus **104** to the user. In the conventional operation, the user inputs the user's e-mail address on the setting screen of

the data processing device **101**. In this regard, in the present exemplary embodiment, the user instructs the image forming apparatus **104** to use the user's e-mail address preliminarily registered in the image forming apparatus **104** on the data processing device **101**. Accordingly, the user can instruct to send image data to the user without setting the user's e-mail address on the data processing device **101**.

[0058] A description will be given of a setting that is performed on the data processing device **101** to send scanned image to the user with reference to FIGS. **8**, **9**, and **10**.

[0059] The send-to-myself setting screen **710** illustrated in FIG. **8** is a screen to perform a setting to send scanned image to the user by e-mail, and the screen is displayed by the send-to-myself button **701** being pressed on the screen illustrated in FIG. **7**. In a title area **711**, a title indicating what kind of a setting is to be performed on the screen ("send e-mail to myself" in FIG. **8**) is displayed, and a button to be used to cancel the setting and go back to the menu screen is also disposed. An e-mail address setting area **712** is an area to display the set e-mail address. In a case where the area is pressed, the e-mail address as the sending destination can be directly input by software keys. In a case where an address book open button **713** is pressed, an address book in the data processing device **101** opens. In a case where the user selects an e-mail address from the address book that has opened, the selected e-mail address is displayed in the e-mail address setting area **712**. A plurality of e-mail addresses can be set. In a send-to-myself-setting-in-MFP setting area **714**, a switching button **715** is used to switch whether or not to cause the image forming apparatus **104** to execute a function corresponding to that of the send-to-myself button **602** of a multi-function peripheral (MFP) (as an example of the image forming apparatus **104**). That is, whether or not to use the user's e-mail address registered in the image forming apparatus **104** can be switched by the switching button **715**. In the present exemplary embodiment, the switching button **715** in the send-to-myself-setting-in-MFP setting area **714** is ON by default. In a case where an open button **718** is pressed, the screen transits to a screen illustrated in FIG. **9**, and settings regarding a subject of an e-mail, a body, and the like can be set.

[0060] A CC input area **720** and a BCC input area **721** each have the same configuration of that of the e-mail address setting area **712**, and an e-mail address to which an e-mail is sent can be input thereto. A subject field **722** to which a subject is input, and a file name field **723**, a body field **724** to which a body is input can be input by software keys of the data processing device **101**, and the user can input any character strings thereto.

[0061] The description returns to FIG. **8**. Buttons, for changing the scan setting, including a color setting button **717** are disposed below a scan setting area **716**. In a case where the color setting button **717** is pressed, the screen transits to a color setting screen **730** illustrated in FIG. **10**. In a title area **731**, a title indicating that the screen is the "color setting" screen is displayed, and a back button to confirm the setting and return to the previous screen is disposed. Setting items **732** display items regarding the color setting. In a case where any one of these items is pressed, the item is changed to a selected status as illustrated in a status **733**.

[0062] After various kinds of settings are performed on the send-to-myself setting screen **710**, and in a case where the user brings the data processing device **101** in touch with the image forming apparatus **104**, the data processing device **101** sends the settings performed on the send-to-myself setting screen **710** to the image forming apparatus **104**. The image forming apparatus **104** then executes sending processing based on the received settings.

[0063] A save-to-my-folder setting screen (not illustrated) displayed by the save-to-my-folder button **702** illustrated in FIG. **7** being pressed has a configuration equivalent to the send-to-myself setting screen illustrated in FIG. **8**. The send-to-myself-setting-in-MFP setting area **714** illustrated in FIG. **8** is replaced with an area for determining whether to execute a function corresponding to that of the save-to-my-folder button **603**. In a case where the setting is ON, the user's folder preliminarily registered in the image forming apparatus **104** is set to a sending destination of data.

[0064] A description will be given of the processing procedure of the data processing device **101** in

a case where various kinds of settings are performed on the send-to-myself setting screen **710** and the user brings the data processing device **101** in touch with the image forming apparatus **104**, using a flowchart for sending illustrated in FIG. **11**. Each step illustrated in the flowchart is processed by the CPU **202** loading a control program (application) stored in a memory, such as the ROM **203** and the storage device **209**, to the RAM **204** and executing the program.

[0065] In step **S801**, the send-to-myself button **701** on the menu screen **703** illustrated in FIG. **7** is pressed, and the UI control unit **402** displays the send-to-myself setting screen **710**. In step **802**, the CPU **202** accepts a setting regarding a sending destination. Specifically, the CPU **202** accepts an e-mail address setting performed in the e-mail address setting area **712** and switching by the switching button **715** of whether or not to execute the send-to-myself operation of the MFP. In step **S803**, the CPU **202** accepts an input regarding e-mail information, such as a scan setting performed in the scan setting area **716**, the subject field **722**, the file name field **723**, and the body field **724**. Since, in the present exemplary embodiment, the switching button **715** is ON by default and also a default value is set to the scan setting, steps **S802** and **S803** are not necessarily required, and the order of steps are not limited to this order. That is, the processing can proceed to the next step without the processing of accepting the sending destination, the scan setting, and the like.

[0066] In step **S804**, after the user brings the data processing device **101** in touch with the image forming apparatus **104**, the device connection unit **404** acquires connection information on wireless LAN communication from an NFC tag of the image forming apparatus **104**. In step **S805**, the CPU **202** determines whether the data processing device **101** has established communication connection with the image forming apparatus **104** using the connection information. In a case where the data processing device **101** has not established the communication connection (NO in step **S805**), the UI control unit **402** displays a message indicating a failure to communicate on the data processing device **101**, and the processing ends. In a case where the data processing device **101** has established the communication connection (YES in step **S805**), the processing proceeds to step **S806**. In step **S806**, the CPU **202** determines whether the setting in the send-to-myself-setting-in MFP setting area **714** is ON or OFF. In a case where the setting in the send-to-myself-setting-in-MFP setting area **714** is ON (YES in step **S806**), the processing proceeds to step **S807**. In step **S807**, the CPU **202** sends authentication information registered in the authentication setting management unit **403** to the image forming apparatus **104**, and performs log-in processing to the image forming apparatus **104**. The communication established in step **S805** is used for the sending in this operation. In step **S808**, the CPU **202** receives a result of the log-in processing from the image forming apparatus **104** and determines whether the log-in is successful. In a case where the log-in is successful (YES in step **S808**), the processing proceeds to step **S809**. In step **S809**, the network control unit **405** sends job setting information (an e-mail address, a send-to-myself setting in the MFP, a scan setting, e-mail information, and the like) set on the send-to-myself setting screen **710** and job information including an execution command to the image forming apparatus **104**. The communication established in step **S805** is used for the sending in this operation. In a case where the network control unit **405** receives a message that the log-in is failed, the processing proceeds to step **S810**. In step **S810**, the UI control unit **402** displays an authentication error on the screen, and the processing ends.

[0067] In a case where the setting in the send-to-myself-setting-in-MFP setting area **714** is OFF (NO in step **S806**), the processing proceeds to step **S811**. In step **S811**, the CPU **202** determines whether an e-mail address is set in the e-mail address setting area **712**. In a case where the e-mail address is set in the e-mail address setting area **712** (YES in step **S811**), the processing proceeds to step **S807**. In a case where no e-mail address is set in the e-mail address setting area **712** (NO in step **S811**), the processing proceeds to step **S812**. In step **S812**, the UI control unit **402** displays an error message that “no destination is set” on the screen, and the processing ends. In this flowchart, the data processing device **101** acquires the connection information from the image forming apparatus **104** via the NFC communication and establishes wireless LAN communication with the

image forming apparatus **104** using the connection information to exchange various kinds of data with the image forming apparatus **104**. However, the configuration is not limited thereto, and the data processing device **101** and the image forming apparatus **104** may exchange various kinds of data via Bluetooth® communication. Alternatively, the data processing device **101** may acquire connection information by reading a quick response (QR) code displayed by the UI control unit **502** of the image forming apparatus **104**, and perform wireless communication using the connection information.

[0068] Subsequently, a description will be given of processing executed by the image forming apparatus **104** that has received the job information (the job information sent in step **S809**) including the job setting information set on the send-to-myself setting screen **710** on the data processing device **101**, using the flowchart illustrated in FIG. **12**. In this operation, the image forming apparatus **104** has received the authentication information sent from the data processing device **101** in step **S807**, and is in a state where the user of the data processing device **101** has logged in to the image forming apparatus **104**. The image forming apparatus **104**, which has received the authentication information and performed the log-in processing, may display a screen after the log-in.

[0069] Each step illustrated in the flowchart is processed by the CPU **302** loading a control program (application) stored in a memory, such as the ROM **304** and the HDD **305**, to the RAM **303** and executing the program. In step **S821**, the network control unit **505** receives the job information sent in step **S809** from the data processing device **101**. In this processing, the CPU **302** may start an application to scan and send data, and the UI control unit **502** may display the screen **601**. In step **S822**, the CPU **302** refers to the send-to-myself-setting-in-MFP setting in the received job information to determine whether the setting is ON or OFF. In a case where the setting is ON (YES in step **S822**), the processing proceeds to step **S823**. In step **S823**, the CPU **302** determines whether an e-mail address of the user who has logged in to the image forming apparatus **104** (for example, a personal e-mail address set in the user's personal computer (PC)) is registered in the image forming apparatus **104**. In a case where the e-mail address has been registered (YES in step **S823**), the processing proceeds to step **S824**. In step **S824**, the CPU **302** adds the user's registered e-mail address to a sending destination list. In a case where the e-mail address has not been registered (NO in step **S823**), the processing proceeds to step **S825**.

[0070] In a case where the "send-to-myself" setting in the MFP is OFF (NO in step **S822**), the processing proceeds to step **S825**. In step **S825**, the CPU **302** extracts a list of e-mail address(es) as sending destination(s) included in the received job information (e-mail address(es) set in the e-mail address setting area **712** and the like), and adds the e-mail address(es) to the sending destination list. In step **S826**, the CPU **302** determines whether the number of e-mail addresses in the sending destination list is one or more. In a case where the number of e-mail addresses is one or more (YES in step **S826**), the processing proceeds to step **S827**. In step **S827**, the CPU **302** extracts a scan setting from the job information, and executes scan with the scan setting. In step **S828**, after the execution of the job is completed, the CPU **302** creates a sending e-mail including e-mail information. In this operation, the CPU **302** sets an e-mail address in the sending destination list as a destination. In step **S829**, the CPU **302** adds the scanned image scanned in step **S827** to the e-mail created in step **S828** and sends the email.

[0071] In a case where there is no e-mail address in the sending destination list in step **S826** (NO in step **S826**), the processing proceeds to step **S830**. In step **S830**, the UI control unit **502** displays an error message that no destination is set, and the processing ends. In this operation, the error message may be displayed on the data processing device **101**.

[0072] As described above, according to the present exemplary embodiment, the sending of the scanned image to the user's e-mail address registered in the image forming apparatus can be instructed from the data processing device (mobile terminal). This eliminates the need for the user inputting the user's e-mail address on the data processing device **101**, and therefore usability is

increased.

[0073] For example, there is a case where the user's e-mail address for work is not registered in an address book of the mobile terminal of the user. Meanwhile, user ID of each user who can use the image forming apparatus and each user's e-mail address for work are registered in the image forming apparatus by an administrator of the image forming apparatus. Under such circumstances, according to the present exemplary embodiment, the user can easily issue an instruction to send scanned image to the user's e-mail address for work without inputting the user's e-mail address for work on the mobile terminal.

[0074] While, in the present exemplary embodiment, settings are performed on the data processing device and then the data processing device is brought in touch with the image forming apparatus to start communication, the configuration is not limited thereto. The present exemplary embodiment is also applicable to a case where the data processing device and the image forming apparatus are in communication with each other, and the settings are performed on the data processing device and an instruction is then issued to the image forming apparatus. While, in the present exemplary embodiment, the description has been given of the example of using the function of the "send-to-myself" setting in the MFP on the data processing device, a case of using the save-to-my-folder setting in the MFP on the data processing device can be achieved by a similar configuration.

[0075] In the first exemplary embodiment, in a case where there is no e-mail address in the sending destination list in step **S826** for such reasons as no e-mail address as a destination being set on the data processing device **101**, the error message is displayed in step **S830**. In a second exemplary embodiment, a description will be given of an example in which the data processing device **101** automatically acquires a personal e-mail address of the data processing device **101**, and sets the personal e-mail address as a destination of the send-to-myself operation. Since the basic configuration of the present exemplary embodiment is the same as that of the first exemplary embodiment, only a difference will be described.

[0076] The processing procedure of the image forming apparatus **104** according to the present exemplary embodiment will be described using the flowchart illustrated in FIG. **13**. Since the basic configuration of the flowchart in FIG. **13** is the same as that in FIG. **12**, only a difference will be described. In a case where the CPU **302** of the image forming apparatus **104** performs determination of whether the number of e-mail addresses in the sending destination list is one or more, and then determines that there is no e-mail address in the sending destination list in step **S826**, the processing proceeds to step **S901**. In step **S901**, the CPU **302** notifies the data processing device **101** that no destination is set. In step **S902**, the CPU **302** determines whether a predetermined period of time has elapsed after the notification in step **S901**. In a case where the predetermined period of time has elapsed (YES in step **S902**), the UI control unit **502** displays an error message or the CPU **302** notifies the data processing device **101** of the error, and the processing ends. In a case where the predetermined period of time has not elapsed (NO in step **S902**), the processing proceeds to step **S903**. In step **S903**, the CPU **302** determines whether the image forming apparatus **104** has received a response from the data processing device **101**. In a case where the CPU **302** has received no response (NO in step **S903**), the processing returns to step **S902**. In a case where the CPU **302** has received the response (YES in step **S903**), the processing proceeds to step **S904**. In step **S904**, the CPU **302** determines whether an e-mail address is included in the received response. In a case where the e-mail address is included in the received response (YES in step **S904**), the processing proceeds to step **S905**. In step **S905**, the CPU **302** adds the e-mail address included in the received response to the sending destination list, and then the processing proceeds to step **S827**. In a case where no e-mail address is included in the received response (NO in step **S904**), the CPU **302** cancels the processing and the UI control unit **502** displays an error message, and the processing ends.

[0077] The processing procedure executed by the data processing device **101** to search for the user's personal e-mail address of the data processing device **101**, and send the searched e-mail

address to the image forming apparatus **104** will be described using the flowchart illustrated in FIG. **14**.

[0078] In step **S911**, the CPU **202** of the data processing device **101** receives the notification indicating that no destination is set from the image forming apparatus **104** in step **S901**. In step **S912**, the CPU **202** searches for the user's personal e-mail address of the data processing device that is set to the data processing device **101**. The user's personal e-mail address of the data processing device **101** mentioned herein is, a personal e-mail address set to the operating system of the data processing device **101**, such as a carrier e-mail address and a Gmail address. In step **S913**, the CPU **202** determines whether the number of e-mail addresses searched in step **S912** is one or more. In a case where the number of e-mail addresses is one or more (YES in step **S913**), the processing proceeds to step **S914**. In step **S914**, the UI control unit **402** displays a screen (not illustrated) including a message for making a user's confirmation of the personal e-mail address(es) to be sent to the image forming apparatus **104**, together with OK and cancel buttons. In step **S915**, the CPU **202** determines whether a pressed button is the OK button. In a case where the pressed button is the OK button (YES in step **S915**), the processing proceeds to step **S916**. In step **S916**, the CPU **202** sends the e-mail address(es) searched in step **S912** to the image forming apparatus **104**, and the processing ends. In a case where the pressed button is the cancel button (NO in step **S915**), the processing proceeds to step **S918**. In step **S918**, the CPU **202** notifies the image forming apparatus **104** of cancellation of the job, and the processing ends.

[0079] In a case where no e-mail address can be searched (NO in step **S913**), the processing proceeds to step **S917**. In step **S917**, the UI control unit **402** displays a message that there is no e-mail address on the data processing device **101**, and the processing proceeds to step **S918**.

[0080] While the data processing device **101** displays the confirmation message in step **S914** in this flowchart, the data processing device **101** may send the user's personal e-mail address(es) of the data processing device **101** that can be automatically searched to the image forming apparatus **104** without displaying the confirmation message.

[0081] According to the exemplary embodiments described above, even in a case where the user issues an instruction for the send-to-myself operation to the image forming apparatus **104** without setting an e-mail address as a destination on the data processing device **101**, the user can set the user's personal e-mail address of the data processing device **101** as the destination without the effort of setting the e-mail address. While, in the present exemplary embodiment, the data processing device **101** is configured to search for the user's personal e-mail address of the data processing device **101** in a case where no destination is set at the time of sending a job, the configuration is not limited thereto. The data processing device **101** may be configured such that the searched user's personal e-mail address of the data processing device **101** is set by default in the e-mail address setting area **712**.

[0082] The present disclosure includes the following processing. That is, software (program) to implement the functions of the exemplary embodiments described above is supplied to a system or a device via a network or a storage medium of various types, and the processing is executed by a computer (or a CPU, a microprocessing unit (MPU), or the like) of the system or device reading the program.

OTHER EMBODIMENTS

[0083] Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium

to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0084] While the present disclosure includes exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A method for a system, the system including devices and an image forming apparatus, the devices including an information processing device, the image forming apparatus including a display unit, the method comprising: displaying, on a display unit of the information processing device, an area for setting authentication information of a user who logs in to the image forming apparatus; registering, based on a user operation on the area for setting the authentication information, the authentication information into the information processing device; transmitting, based on a user operation at the information processing device for causing the image forming apparatus to execute an image forming function, the registered authentication information and job information by the information processing device to the image forming apparatus, wherein the job information is information for causing the image forming apparatus to execute the image forming function; receiving the authentication information and the job information at the image forming apparatus; executing login processing by using the authentication information, without receiving any user operation for executing the login processing on an operation unit of the image forming apparatus, at the image forming apparatus; and executing the image forming function based on the job information, without receiving any user operation for executing the image forming function on the operation unit of the image forming apparatus, at the image forming apparatus.
2. The method according to claim 1, further comprising: displaying a screen of the image forming function corresponding to the job information transmitted to the image forming apparatus, without receiving any user operation for displaying the screen of the image forming function corresponding to the job information transmitted to the image forming apparatus on the operation unit of the image forming apparatus, at the image forming apparatus.
3. The method according to claim 1, further comprising: acquiring connection information from the image forming apparatus via short-range wireless communication by the information processing device; and performing wireless local area network (LAN) communication with the image forming apparatus using the acquired connection information by the information processing device, wherein transmitting includes transmitting the job information and the authentication information to the image forming apparatus using the LAN communication.
4. The method according to claim 3, wherein, after the connection information is acquired, processing until transmitting of the job information is automatically performed.
5. The method according to claim 1, wherein, in a case of failure in the login processing executed using the received authentication information at the image forming apparatus, an error notification is made from the image forming apparatus to the information processing device.
6. The method according to claim 1, wherein the image forming function is processing using a

setting value corresponding to the received authentication information.

7. The method according to claim 1, wherein a scan image obtained by executing scanning based on the job information is transmitted to a destination registered in the image forming apparatus in association with the received authentication information.

8. The method according to claim 1, further comprising: acquiring connection information from the image forming apparatus via short-range wireless communication by the information processing device; and performing wireless local area network (LAN) communication with the image forming apparatus using the acquired connection information by the information processing device, wherein transmitting includes transmitting the job information to the image forming apparatus using the LAN communication.

9. The method according to claim 8, wherein, after the connection information is acquired, processing until transmitting of the job information is automatically performed.

10. The method according to claim 1, wherein the image forming apparatus transmits generated image data to a transmission destination included in the transmitted job information.

11. The method according to claim 1, wherein the image forming apparatus has a scan function.

12. The method according to claim 1, wherein the operations further include receiving a setting of the job information to be transmitted at the information processing device, and wherein the job information is information for transmitting, by the image forming apparatus, an image scanned by the image forming apparatus.

13. The method according to claim 1, wherein the image forming apparatus is configured to transmit generated image data to a destination registered in the image forming apparatus in association with the received authentication information.

14. The method according to claim 1, wherein image data is a scan image obtained by executing scanning based on the job information.

15. The method according to claim 1, further comprising: executing the login processing by using the received authentication information, and, without receiving any user operation for performing screen display on the operation unit of the image forming apparatus, displaying a screen after the login processing on the display unit of the image forming apparatus.

16. A method for an image forming apparatus including a display unit, the method comprising: receiving, based on a user operation on an area for setting authentication information of a user who logs in to the image forming apparatus, the area being displayed on a display unit of an information processing device, authentication information registered in the information processing device and job information; executing login processing by using the authentication information, without receiving any user operation for executing the login processing on an operation unit of the image forming apparatus, at the image forming apparatus; and executing the image forming function based on the job information, without receiving any user operation for executing the image forming function on the operation unit of the image forming apparatus, at the image forming apparatus.

17. An image forming apparatus including a display unit, the image forming apparatus comprising: one or more processors; and at least one memory storing executable instructions, which when executed by the one or more processors, cause the image forming apparatus to perform operations comprising: receiving, based on a user operation on an area for setting authentication information of a user who logs in to the image forming apparatus, the area being displayed on a display unit of an information processing device, authentication information registered in the information processing device and job information; executing login processing by using the authentication information, without receiving any user operation for executing the login processing on an operation unit of the image forming apparatus, at the image forming apparatus; and executing the image forming function based on the job information, without receiving any user operation for executing the image forming function on the operation unit of the image forming apparatus, at the image forming apparatus.
